CS 122 Final Project

Overview

For the final project of CS 122, you will create an interface written in Python to obtain, analyze, and visualize data from the internet. The project will be formulated, worked on, and presented *in pairs*. The overall project (100 points) will be assessed in terms of its components (80 points) and its presentation (20 points), described below.

Components (80 points) – Due by 1 pm on 12/10

- Interface (GUI or Web Interface, 30 points)
  Your interface can be either a Graphic User Interface (GUI) generated with Python's Tkinter library, or a web interface, created with Flask. Your interface should have at least two windows, such as a home screen and a pop-up window. Your interface must have at least 10 "widgets" such as buttons, drop-down windows, or otherwise that allow the user to gather, update, and interact with the data.
- Access Web Data (20 points for partner 1)
  Your program should access data from the internet from a web page with freely available and accessible information.
- Data Organization (20 points for partner 1)
  The data you obtain from the internet should be organized in a file format and structure that matches your data. For example, you may choose to organize your data in a spreadsheet, a set of daily files, etc.
- Data Analysis (20 points for partner 2)
  Your program should conduct an analysis of your data. For example, you may choose to calculate trends in the data using Python's numpy package, statistics in your data from Python's scipy module, or any other analysis that makes sense for your data.
- Visualization (20 points for partner 2)
  Your program should contain a visualization component to generate at least one detailed plot of the data. Your plot can be a time series, a histogram, a map, or any other type of figure that demonstrates your data.
- Github (10 points)
  The code for your program will be hosted on Github and documented with a detailed README.md file with a description of the program, instructions for installation, and a list of updates you would pursue if you were to continue working on your project in another class.

Presentation (20 points) – In Class Week 16 (12/1 and 12/3)

During week 16, you and your partner will present your program to the class. Your presentation should be 7 minutes in length split approximately evenly between you and your partner (i.e. about 3.5 minutes each). The presentation should focus on how users interact with the program rather than the structure of the underlying code. You should demonstrate where you obtain your data and what analysis was done on your data to create your visualization.

Grading

While you will be working on this project with a partner, your effort will be assessed on an individual basis. Each person will be responsible for one page of the interface. The responsibilities of the underlying code will be split between each partner: one person will be responsible for gathering your data from the web and organizing it on the local computer; the other person will be responsible for the analysis and visualization of the data.

Examples

- Interface to obtain weather data, analyze anomalies, and plot a forecast
- Interface to download stock market data, analyze market growth or loss, and plot the trends
- Interface to obtain sports scores for a tournament, fill out a bracket, and assess which teams are most likely to win
- Interface to gather campsite availability data for a local campground and plot the availability on a map with a quantification for how popular a particular site may be